

RCC BEAM DESIGN REPORT - B1

IS 456:2000 Limit State Design | Generated 2026-08-28

1. Design Parameters

Parameter	Value	Unit
Span	5.00	m
Width, b	300	mm
Overall Depth, D	500	mm
Clear Cover	25	mm
Support Condition	Simply Supported	-
Loading	Uniformly Distributed	-

2. Material Properties

Parameter	Value	Unit
Concrete Grade	M25	-
Steel Grade	Fe415	-
Stirrup Diameter	8	mm
Stirrup Legs	2	-

3. Load Calculation

Parameter	Value	Unit
Dead Load (excl. self-weight)	12.00	kN/m
Live Load	8.00	kN/m
Self-weight	3.75	kN/m
Total Service Load	23.75	kN/m
Factored Load, w_u ($1.5 \times DL+LL$)	35.62	kN/m

4. Bending Moment and Shear Force

Parameter	Value	Unit
Design Moment, $M_u = w_u.L^2/8$	111.33	kNm
Design Shear, $V_u = w_u.L/2$	89.06	kN
Effective Depth, d	459.0	mm

6. Flexural Design

Parameter	Value	Unit
Mu,lim (Annex G-1.1(c))	218.00	kNm
Section Type	Singly Reinforced	-
Ast, required (Annex G-1.1(b))	737.7	mm ²
Ast, min (Cl. 26.5.1.1(a))	282.0	mm ²
Ast, max (Cl. 26.5.1.1(b))	6000.0	mm ²

[OK] Flexure: SAFE

7. Shear Design

Parameter	Value	Unit
Nominal Shear Stress, tau_v (Cl. 40.1)	0.647	N/mm ²
Design Shear Strength, tau_c (Table 19)	0.517	N/mm ²
Maximum Shear Stress, tau_c,max (Table 20)	3.10	N/mm ²
Reinforcement Basis	Designed	-
Stirrups Spacing Provided	300	mm c/c

[OK] Shear: SAFE

8. Reinforcement Details

Parameter	Value	Unit
Main Reinforcement	4 x 16	mm
Area Provided, Ast,provided	804.2	mm ²
Excess Steel	9.0	%
Clear Bar Spacing (Cl. 26.3.2)	56.7	mm
Stirrups	2-legged 8	mm @ 300 mm c/c
Development Length, Ld (Cl. 26.2.1)	645	mm

9. Serviceability Checks

Parameter	Value	Unit
Basic Span/Depth Ratio (Cl. 23.2.1)	20.0	-
Modification Factor, kt (Fig. 4)	0.924	-
Modification Factor, kc (Fig. 5)	1.00	-
Allowable Span/Depth Ratio	18.49	-
Actual Span/Depth Ratio	10.89	-

[OK] Serviceability: SAFE

10. IS 456 Compliance Summary

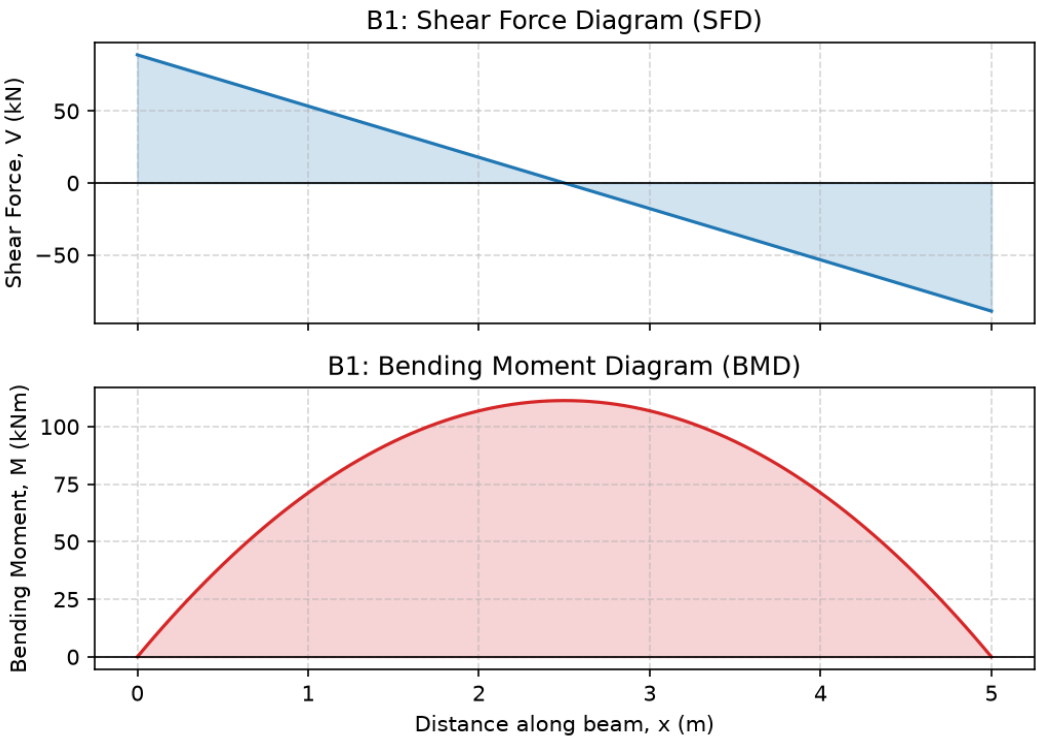
Check	Clause	Status
Load Factor	Table 18 / Cl. 36.4.1	Applied
Flexural Design	Annex G	SAFE
Minimum/Maximum Reinforcement	Cl. 26.5.1.1	SAFE
Shear Design	Cl. 40	SAFE
Bar Spacing	Cl. 26.3.2	OK
Development Length	Cl. 26.2.1	Checked (not a pass/fail limit)
Deflection (Span/Depth)	Cl. 23.2.1	SAFE

11. Final Design Summary

Parameter	Value	Unit
Beam	B1	-
Span	5.0	m
Section	300 x 500	mm
Concrete / Steel	M25 / Fe415	-
Factored Load	35.62	kN/m
Design Moment, Mu	111.33	kNm
Design Shear, Vu	89.06	kN
Main Reinforcement	4 x 16 mm	-
Stirrups	2-legged 8 mm @ 300 mm c/c	-

[OK] Flexure: SAFE
[OK] Shear: SAFE
[OK] Serviceability: SAFE
[OK] OVERALL DESIGN STATUS: SAFE

12. Shear Force and Bending Moment Diagrams



Note: This report is a design-calculation summary generated for engineering review. It is not a substitute for a licensed structural engineer's sign-off, and does not yet include ductile-detailing (IS 13920) or crack-width checks. A reinforcement detailing sketch is planned for a future version of this tool.